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Fuzz4All: Universal Fuzzing with Large Language Models

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Fuzz4All: Universal Fuzzing with Large Language Models

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ABSTRACT

Fuzzing has achieved tremendous success in discovering bugs and vulnerabilities in various software systems. Systems under test (SUTs) that take in programming or formal language as inputs, e.g., compilers, runtime engines, constraint solvers, and software libraries with accessible APIs, are especially important as they are fundamental building blocks of software development. However, existing fuzzers for such systems often target a specific language, and thus cannot be easily applied to other languages or even other versions of the same language. Moreover, the inputs generated by existing fuzzers are often limited to specific features of the input language, and thus can hardly reveal bugs related to other or new features. This paper presents Fuzz4All, the first fuzzer that is *universal* in the sense that it can target many different input languages and many different features of these languages. The key idea behind Fuzz4All is to leverage large language models (LLMs) as an input generation and mutation engine, which enables the approach to produce diverse and realistic inputs for any practically relevant language. To realize this potential, we present a novel autoprompting technique, which creates LLM prompts that are well-suited for fuzzing, and a novel LLM-powered fuzzing loop, which iteratively updates the prompt to create new fuzzing inputs. We evaluate Fuzz4All on nine systems under test that take in six different languages (C, C++, Go, SMT2, Java, and Python) as inputs. The evaluation shows, across all six languages, that universal fuzzing achieves higher coverage than existing, language-specific fuzzers. Furthermore, Fuzz4All has identified 98 bugs in widely used systems, such as GCC, Clang, Z3, CVC5, OpenJDK, and the Qiskit quantum computing platform, with 64 bugs already confirmed by developers as previously unknown.

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1 INTRODUCTION

Fuzz testing [69, 84], also known as fuzzing, is an automated testing approach for generating inputs designed to expose unexpected behaviors, e.g., crashes, of a system under test (SUT). Researchers and practitioners have successfully built practical fuzzing tools, which have shown great success in finding numerous bugs and vulnerabilities in real-world systems [6]. A particularly important family of SUTs are systems that take in programming or formal language inputs, e.g., compilers, runtime engines, and constraint solvers. Numerous fuzzers have been proposed for such systems since they are the fundamental building blocks for software development [12]. For example, finding bugs in compilers and runtime engines is crucial because they can affect all corresponding downstream applications.

Traditional fuzzers can be categorized into generation-based [34, 49, 81] and mutation-based [21, 31, 69]. Generation-based fuzzers aim to directly synthesize complete code snippets, e.g., using a predefined grammar for the target language. Instead of synthesizing from scratch, mutation-based fuzzers apply mutation operators or transformation rules to a set of high quality fuzzing seeds. Unfortunately, both traditional fuzzing approaches face the following limitations and challenges:

C1: Tight coupling with target system and language. Traditional fuzzers are often designed to target a specific language or a particular SUT. However, designing and implementing a fuzzer is extremely time-consuming. For example, Csmith [81], a fuzzer for C/C++ compilers, has more than 80k lines of code, while Syzkaller [70], a fuzzer for Linux system calls, contains tens of thousands of handcrafted rules [10] to generate and modify system calls. Because each target language is different, it is often non-trivial to reuse the effort of implementing a fuzzer from one input language for another. Furthermore, fuzzing strategies that work well for one SUT may not work at all for another one.

C2: Lack of support for evolution. Real-world systems are constantly evolving, e.g., by adding new features to the input language. Traditional fuzzers designed for a specific version of a language or SUT may lose their effectiveness on a new version and cannot be easily used to test newly implemented features. For example, Csmith supports only a limited set of features up to C++11, while the C++ language has evolved significantly since then. In fact, recent work [20] shows that over a six-month fuzzing period, Csmith was not able to uncover any new bugs in the latest releases of the

GCC and Clang compilers, showing that new versions of compilers are becoming immune to existing fuzzers.

C3: Restricted generation ability. Even within the scope of a specific target language, both generation-based and mutation-based fuzzing often are unable to cover a large part the input space. Generation-based fuzzers heavily rely on an input grammar to synthesize valid code, and additionally are equipped with semantic rules that ensure the validity of the synthesized code. To generate a high amount of valid fuzzing inputs or to side-step difficult-to-model language features, generation-based fuzzers often use a subset of the full language grammar, which limits them to test only a subset of all language features. Similarly, mutation-based fuzzers are limited by their mutation operators and require high quality seeds that can be difficult to obtain.

Our work. We present Fuzz4All, the first fuzzer that is *universal* in the sense that it can target many different input languages and many different features of theses languages. Our approach fundamentally differs from existing general-purpose fuzzers, e.g., AFL [50] and libFuzzer [43], which use extremely simple mutations, are unaware of the target language, and therefore struggle to produce meaningful programming language fuzzing inputs. Instead, our key idea is to leverage a large language model (LLM) as an input generation and mutation engine. Because LLMs are pre-trained on large amounts of examples in various programming languages and other formal languages, they come with an implicit understanding of the syntax and semantics of these languages. Fuzz4All leverages this ability by using an LLM as a universal input generation and mutation engine.

The input to Fuzz4All are user-provided documents describing the SUT, and optionally, specific features of the SUT to focus on, e.g., in the form of documentation, example code, or formal specifications. However, these user inputs may be too verbose to directly use as a prompt for the LLM. Instead of requiring the user to manually engineer a prompt [47], which is time-consuming, we present an *autoprompting* step that automatically distills all user-provided inputs into a concise and effective prompt for fuzzing. This prompt is the initial input to an LLM that generates fuzzing inputs. Since continuously sampling with the same prompt would lead to many similar fuzzing inputs, we present an *LLM-powered fuzzing loop*, which iteratively updates the prompt to generate a diverse set of fuzzing inputs. To this end, Fuzz4All combines fuzzing inputs generated in previous iterations with natural language instructions, e.g., asking to mutate these inputs. The LLM-generated fuzzing inputs are then passed to the SUT, which we validate against a user-provided test oracle, such as checking for system crashes.

Fuzz4All addresses the previously discussed limitations and challenges of traditional fuzzers. Instead of meticulously designing a single-purpose fuzzer for a specific SUT (C1), Fuzz4All, by using an LLM as the generation engine, can be applied to a wide range of SUTs and input languages. Compared to existing fuzzers that target a specific version of the SUT or input language (C2), Fuzz4All can easily evolve with the target. For example, to fuzz-test a newly implemented feature, a user can simply provide documentation or example code related to that feature. To address the restricted generation ability of traditional fuzzers (C3), Fuzz4All exploits the fact that LLMs are pre-trained on billions of code snippets, enabling them to create a wide range of examples that likely obey the syntactic and semantic constraints of the input language. Finally, Fuzz4All does not require any instrumentation of the SUT, making the approach easily applicable in practice.

We perform an extensive evaluation on six input languages (C, C++, SMT, Go, Java, and Python) and nine SUTs. For each of them, we compare our approach against state-of-the-art generation-based and mutation-based fuzzers. The results show that Fuzz4All achieves the highest code coverage across all languages, improving the previous state-of-the-art coverage by 36.8%, on average. Additionally, we demonstrate that Fuzz4All supports both general fuzzing and fuzzing targeted at specific features of the SUT, which a user decides upon by providing adequate input documents. Finally, Fuzz4All detects 98 bugs across our studied SUTs, with 64 already confirmed by developers as previously unknown.

Contributions: This paper makes the following contributions:

- ⋆ **Universal fuzzing**. We introduce a new dimension for fuzzing that directly leverages the multi-lingual capabilities of LLMs to fuzz-test many SUTs with a wide range of meaningful inputs.
- ⋆ **Autoprompting for fuzzing**. We present a novel autoprompting stage to support both general and targeted fuzzing by automatically distilling user inputs into a prompt that is effective at generating inputs to the SUT.
- ⋆ **LLM-powered fuzzing loop**. We present an algorithm that continuously generates new fuzzing inputs by iteratively modifying the prompt with selected examples and generation strategies.
- ⋆ **Evidence of real-world effectiveness**. We show across six popular languages and nine real-world SUTs (e.g., GCC, CVC5, Go, javac, and Qiskit) that our approach significantly improves coverage compared to state-of-the-art fuzzers (avg. 36.8%) and detects 98 bugs, with 64 already confirmed as previously unknown.

2 BACKGROUND AND RELATED WORK

2.1 Large Language Models

Recent developments in natural language processing (NLP) has lead to the wide-spread adoption of large language models (LLMs) for both natural language [8] and code tasks [80]. State-of-the-art LLMs are based on transformers [73] and can be classified into decoder-only (e.g., GPT3 [8] and StarCoder [41]), encoder-only (e.g., BERT [19] and CodeBERT [22]) and encoder-decoder (BART [40] and CodeT5 [83]) models. More recently, instruction-based LLMs (e.g., ChatGPT [65] and GPT4 [55]) and LLMs fine-tuned using reinforcement learning from human feedback (RLHF) [88] are shown to understand and follow complex instructions [4, 56, 65].

LLMs are typically either fine-tuned [63] or prompted [47] to perform specific tasks. Fine-tuning updates the model weights through further training on a task-specific dataset. However, suitable datasets may be unavailable, and as LLM sizes continue to grow [35], fine-tuning an LLM is also increasingly expensive. Prompting, on the other hand, does not require explicitly updating the model weights, but provides the LLM with a description of the task, and optionally, a few examples of solving the task. The process of picking the input (i.e., prompt) is known as prompt engineering [47], where a user tries different input instructions until finding one that works well. Recently, researchers have proposed *autoprompting* [68], an automatic process that uses LLM gradients to

select either soft prompts [42, 62], i.e., continuous vector embeddings, or hard prompts [64, 71], i.e., natural language text. Even more recently, researchers have substituted gradient-based methods by computing a proxy score of effectiveness [87].

This work leverages LLMs for the important problem of fuzzing. Unlike traditional autoprompting and proxy-based approaches, our autoprompting strategy directly synthesizes prompts using GPT4 and scores them according to a fuzzing-specific goal.

2.2 Fuzzing and Testing

Fuzz testing aims to generate inputs that cause unexpected behaviors of the SUT. Traditional fuzzers can be classified into generation-based [34, 49, 81] and mutation-based [21, 31, 69]. Generation-based fuzzers create complete code snippets using pre-defined grammars and built-in knowledge of the semantics of the target language. Csmith [81] and YARPGen [49] hard-code language specifications to ensure the validity of generated code snippets to test C and C++ compilers, respectively. jsfunfuzz [34] combines a language grammar with historical bug-triggering code snippets to generate new inputs to test JavaScript engines. Generation-based fuzzers have also been used to test OpenCL [44], the JVM [11], CUDA [33], deep learning compilers [45], Datalog engines [53], and interactive debuggers [38]. Mutation-based fuzzers [69] iteratively perform transformations on seeds to generate new fuzzing inputs. In addition to basic mutations, researchers have developed complex transformations targeted at ensuring type consistency [11, 59], adding historical bug-triggering code snippets [31, 86], and coverage feedback [3, 21, 46]. To benefit from both generation and mutation, many fuzzers use a combination of both approaches [12, 51].

Different from the above fuzzers, which target specific SUTs or languages, another line of research is on general-purpose fuzzing. AFL [50] and libFuzzer [43] are general-purpose fuzzers that use genetic algorithms with a fitness function to prioritize fuzzing inputs for further mutations that achieve new coverage. These mutations are unaware of the SUT and focus on byte-level transformations. That is, when applied on SUTs that receive programming languages as input, general-purpose fuzzers are extremely unlikely to produce valid inputs. Recent work [28] has instead added regular expression-based mutation operators to match common programming statements (e.g., change + to -). The simplicity of these mutation operators limits the ability of such fuzzers at covering new code, especially in more complex languages, such as C [21, 28]. PolyGlot [14] is another language-agnostic fuzzer, which first parses the seed programs into a uniform intermediate representation using a language-specific grammar and then uses a set of mutation operators to generate new programs. While promising, PolyGlot still uses a limited set of mutations and cannot achieve the same level of coverage as fuzzers that are designed for a particular language [21].

To complement traditional fuzzing techniques and apply fuzzing to emerging domains, learning-based fuzzers have been proposed. Prior learning-based techniques mainly focus on training a neural network to generate fuzzing inputs. TreeFuzz [60] parses the training corpus into a tree structure and through tree traversal, learns a probabilistic, generative model that synthesizes new fuzzing inputs. Deep learning models have been used to fuzz PDF parsers [26], OpenCL [17], C [48], network protocols [85], and JavaScript [37]. Very recently, researchers have also directly leveraged LLMs for fuzzing specific libraries, e.g., TitanFuzz [18] uses Codex [13] to generate seed programs and InCoder [24] to perform template-based mutation for fuzzing deep learning libraries [61, 72].

Unlike prior learning- and LLM-based fuzzers, Fuzz4All is easily applicable across many programming languages. Prior work trains language-specific models or requires language-specific parsing. Even TitanFuzz, a recent LLM-based approach, is designed specifically for deep learning libraries with hand-crafted prompts and mutation patterns, and therefore cannot be easily extended to other SUTs. Furthermore, unlike existing techniques, which produce general fuzzing inputs in a particular language, Fuzz4All additionally supports targeted fuzzing, which can generate code snippets that focus on selected features.

In addition to fuzzing, LLMs have also been applied to the related problem of unit test generation [5, 39, 54, 66, 74, 82]. CodaMosa [39] interleaves traditional search-based software testing with querying Codex to generate new unit tests whenever a coverage plateau is reached. TestPilot [66] prompts Codex with method source code and example usages to generate unit tests and to fix incorrectly generated tests. In contrast to these LLM-based test generators, which require a specific type of input (e.g., function source code) and only work for unit testing [54, 66], by using our novel autoprompting stage, Fuzz4All can take inputs in arbitrary formats for both general and targeted fuzzing. Furthermore, such unit test generators often require manual work to check or complete the tests as they are limited by automatically generated test-oracles, which even state-of-the-art LLMs [15, 65] cannot always produce reliably. Instead, Fuzz4All leverages widely-used fuzzing oracles, such as crashes, and is fully automated.

3 FUZZ4ALL APPROACH

We present Fuzz4All, a universal fuzzer that leverages LLMs to support both general and targeted fuzzing of any SUTs that take in programming language input. Figure 1 provides an overview of our approach. Fuzz4All first takes in arbitrary *user input* that describes the fuzzing inputs to be generated, e.g., documentation of the SUT, example code snippets, or specifications. As the user input may be long, redundant, and partially irrelevant, the approach distills it into a concise but informative prompt for fuzzing. To this end, Fuzz4All performs an *autoprompting* step (Section 3.1) by using a large, state-of-the-art *distillation LLM* to sample multiple different candidate prompts ❶. Each candidate prompt is passed on to the *generation LLM* to generate code snippets (i.e., fuzzing inputs) ❷. Fuzz4All then selects the prompt that produces the highest quality fuzzing inputs ❸.

Fuzz4All builds on two models, a distillation LLM that reduces the given user input and a generation LLM that creates the fuzzing inputs, to balance the trade-off between the costs and benefits different LLMs provide. Because the distillation LLM needs to understand and distill arbitrary user input, we use a high-end, large foundational model with strong natural language understanding abilities. However, directly using such a large model for input generation would be inefficient due to the high inference cost of autoregressive generation. Instead, to perform efficient fuzzing, Fuzz4All uses a smaller model as the generation LLM. While our approach is general

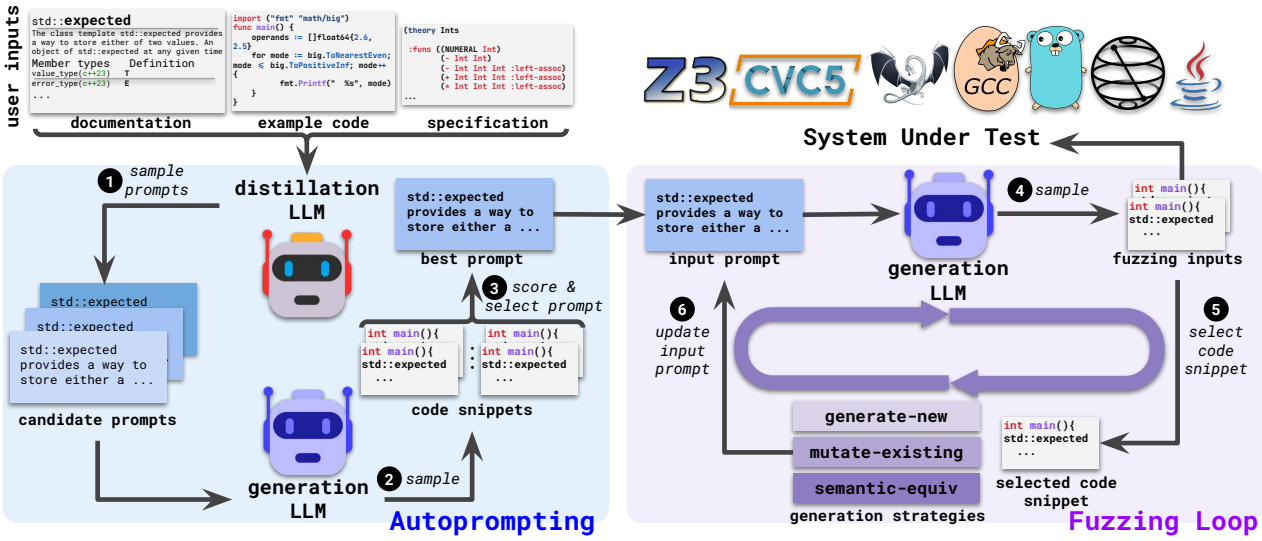


Figure 1: Overview of Fuzz4All.

Algorithm 1: Autoprompting for fuzzing

1 **Function** Autoprompting:
  Input : userInput, numSamples
  Output: inputPrompt
2 greedyPrompt $\leftarrow \mathcal{M}_{\mathcal{D}}$ (userInput, APInstruction, temp=0)
3 candidatePrompts $\leftarrow$ [greedyPrompt]
4 **while** |candidatePrompts| < numSamples **do**
5 prompt $\leftarrow \mathcal{M}_{\mathcal{D}}$ (userInput, APInstruction, temp=1)
6 candidatePrompts $\leftarrow$ candidatePrompts + [prompt]
7 inputPrompt $\leftarrow \underset{p \in \text{candidatePrompts}}{\arg\max}$ Scoring ($\mathcal{M}_{\mathcal{G}}$ (p), SUT)
8 **return** inputPrompt

across any pairs of distillation and generation LLMs, we implement Fuzz4All with the state-of-the-art GPT4 [55] and StarCoder [41].

Using the best prompt selected via autoprompting as the initial input prompt for the generation LLM, we then move on to the *fuzzing loop* (Section 3.2), where Fuzz4All continuously samples the generation LLM to generate fuzzing inputs (4). To avoid generating many similar fuzzing inputs, Fuzz4All continuously updates the input prompt in each iteration. Specifically, the approach selects a previously generated input as an *example* (5), which demonstrates the kind of future inputs we want the model to generate. In addition to the example, Fuzz4All also appends a *generation instruction* to the initial prompt, which guides the model toward generating new fuzzing inputs (6). This process is repeated while continuously passing the generated fuzzing inputs into the SUT and checking its behavior against a user-defined oracle, such as crashes.

3.1 Autoprompting

The following presents the details of the first of two main steps of Fuzz4All, which distills the given user input via autoprompting into a prompt suitable for fuzzing. The user input may describe the SUT in general, or particular feature of the SUT to be tested. As shown in Figure 1, user inputs may include technical documentation, example code, specifications, or even combinations of different modalities. Unlike traditional fuzzers that require inputs to follow a specific format, e.g., code snippets to use as seeds or well-formed specifications, Fuzz4All can directly understand the natural language descriptions or code examples in the user input. However, some information in the user input may be redundant or irrelevant, and hence, directly using the user inputs as a prompt for the generation LLM may be ineffective, as confirmed by our ablation study in Section 5.3. Therefore, the goal of autoprompting is to generate a distilled input prompt that enables effective LLM-based fuzzing.

3.1.1 Autoprompting Algorithm. Algorithm 1 details Fuzz4All's autoprompting step. The inputs are the user input and the number of candidate prompts to generate. The final output is the input prompt selected to be used for the fuzzing campaign. As our goal is to use a distillation LLM to generate prompts that distill the information provided by the user, we give the following autoprompting instruction to the distillation LLM: "Please summarize the above information in a concise manner to describe the usage and functionality of the target". Let $\mathcal{M}_{\mathcal{D}}$ be the distillation LLM, `userInput` be the user input and `APInstruction` be the autoprompting instruction. The prompt `prompt` generated can be formalized as the conditional probability: $\mathcal{M}_{\mathcal{D}}(\texttt{prompt} \mid \texttt{userInput}, \texttt{APInstruction})$

Fuzz4All first generates a candidate prompt using greedy sampling with temperature 0 (line 2). By first sampling with low temperature, the algorithm obtains a plausible solution with a high degree of confidence. This approach is commonly used in other domains, e.g., program synthesis [13], where the greedy output is evaluated first to check if it can solve the problem. The algorithm then moves on to sampling with higher temperature to obtain more diverse prompts (line 5), as done in prior work [13, 79]. Compared to a greedy approach, sampling with high temperature yields different prompts that can each provide a unique distilled summary of the user input. Each generated prompt is added to a list of candidate

High level description of feature

The C++23 std::expected class template provides a way to store either an expected value of type T or an unexpected value of type E. *It is useful for handling functions that may return an error or a valid result.* The stored value is allocated directly within the storage occupied by the expected object, without dynamic memory allocation.

Descriptions of the inputs

The template parameters are T (the expected value type) and E (the unexpected value type). Both types must meet the Destructible requirements, and certain types are not allowed.

Different usages of target

std::expected provides member functions for construction, destruction, assignment, and accessing the stored values. Observers like operator bool and has_value can be used to check if the object contains an expected value. Functions like value, error, and value_or can be used to access the expected or unexpected values.

Monadic operations like and_then, transform, or_else, and transform_error allow chaining operations on expected values and handling errors in a functional manner.

Modifiers like emplace and swap can be used to construct the expected value in-place or exchange the contents of expected objects. Non-member functions like operator== and swap(std::expected) provide comparison and swapping functionality.

Helper classes like unexpected, bad_expected_access, and unexpect_t are used to represent unexpected values, exceptions, and in-place construction tags for unexpected values in expected objects.

Figure 2: Autoprompting result for `std::expected`.

prompts (line 6), until the algorithm reaches the desired number of candidates.

To pick the best input prompt to be used in the fuzzing step, the algorithm evaluates each candidate prompt by performing a small-scale fuzzing experiment. Specifically, the approach uses each prompt as an input to the generation LLM to produce multiple code snippets per prompt. Fuzz4All then scores the generated code snippets for each prompt based on a scoring function. While the scoring function can be based on a variety of different metrics, e.g., coverage, bug finding, or the complexity of generated fuzzing inputs, to make the approach lightweight and general, our scoring function is the number of unique generated code snippets that are valid, i.e., accepted by the target SUT. This metric is chosen since for fuzzing, we want fuzzing inputs to be valid or close to valid to the logic deep inside the SUT. Let $\mathcal{M}_{\mathcal{G}}$ be the generation LLM, p be a candidate prompt, `isValid` be a function that returns 1 if a generated code c is valid and 0 otherwise. Our default scoring function is defined as: $\sum_{c \in \mathcal{M}_{\mathcal{G}}(p)} [\texttt{isValid}(c,\ \texttt{SUT})]$. Finally, Fuzz4All selects the input prompt with the highest score (line 7) as the initial input prompt to be used for fuzzing. In summary, our autoprompting step combines both prompt generation and scoring, which allows Fuzz4All to automatically generate and select a prompt suitable for the fuzzing target.

3.1.2 Example: Autoprompting. Figure 2 shows an example of an input prompt generated by our autoprompting algorithm. The example is for fuzzing C++ compilers while focusing specifically on `std::expected`, a new feature introduced in C++23. As the user input, we pass the original `cppreference` documentation [2] to Fuzz4All, which spans multiple screen lengths with small tables and verbose descriptions (498 words, 3,262 characters). In contrast, the distilled input prompt created by the autoprompting algorithm provides a more concise natural language description of the targeted feature (214 words, 1,410 characters). The input prompt contains a high-level description of how `std::expected` is to be used. For example, the input prompt contains a concise sentence (highlighted in orange) that summarizes the situations the feature is useful in. Additionally, the input prompt contains descriptions of the inputs, as well as the different usages (i.e., member functions) of the feature. For example, functions `and_then`, `transform`, `or_else`, and `transform_error` have very similar descriptions in the original

Algorithm 2: Fuzzing loop

1 **Function** FuzzingLoop:
  Input : inputPrompt, timeBudget
  Output : bugs
2 genStrats ← [`generate-new`, `mutate-existing`, `semantic-equiv`]
3 fuzzingInputs ← $\mathcal{M}_{\mathcal{G}}$ (inputPrompt + `generate-new`)
4 bugs ← `Oracle` (fuzzingInputs, SUT)
5 **while** timeElapsed < timeBudget **do**
6 example ← *sample* (fuzzingInputs, SUT)
7 instruction ← *sample* (genStrats)
8 fuzzingInputs ← $\mathcal{M}_{\mathcal{G}}$ (inputPrompt + example + instruction)
9 bugs ← bugs + `Oracle` (fuzzingInputs, SUT)
10 **return** bugs

documentation, which is repeated for each function. Instead, in the distilled input prompt, these functions are grouped together in a concise manner that still illustrates how they can be used. Using the distilled input prompt, Fuzz4All can generate fuzzing inputs that effectively target the `std::expected` feature of C++ compilers.

3.1.3 Comparison with Existing Autoprompting Techniques. To the best of our knowledge, we are the first to automatically distill knowledge from arbitrary user inputs for a software engineering task using black-box autoprompting. Compared to prior work on autoprompting in NLP [68] and software engineering [75], which optimize the prompt by accessing model gradients, our autoprompting needs only black-box, sampling access to the distillation LLM. While the use of a scoring function to evaluate each prompt is similar to recent work in NLP [87], our scoring function directly evaluates the prompt on the exact downstream task of generating valid code snippets, instead of using an approximate proxy scoring function.

3.2 Fuzzing Loop

Given the input prompt created in the first step of Fuzz4All, the goal of the fuzzing loop is to generate diverse fuzzing inputs using a generation LLM. However, due to the probabilistic nature of LLMs, sampling multiple times using the same input would produce the same or similar code snippets. For fuzzing, we aim to avoid such repeated inputs and instead want to generate a diverse set of fuzzing inputs that cover new code and discover new bugs. To accomplish this goal, we exploit the ability of LLMs to utilize both examples and natural language instructions to guide the generation.

The high-level idea of the fuzzing loop is to continuously augment the original input prompt by selecting an example fuzzing input from previous iterations and by specifying a generation strategy. The goal of using an example is to demonstrate the kind of code snippet we want the generation LLM to produce. The generation strategies are designed as instructions on what to do with the provided code example. These strategies are inspired by traditional fuzzers, mimicking their ability to synthesize new fuzzing inputs (as in generation-based fuzzers) and to produce variants of previously generated inputs (as in mutation-based fuzzers). Before each new iteration of the fuzzing loop, Fuzz4All appends both an example and a generation strategy to the input prompt, enabling the generation LLM to continuously create new fuzzing inputs.

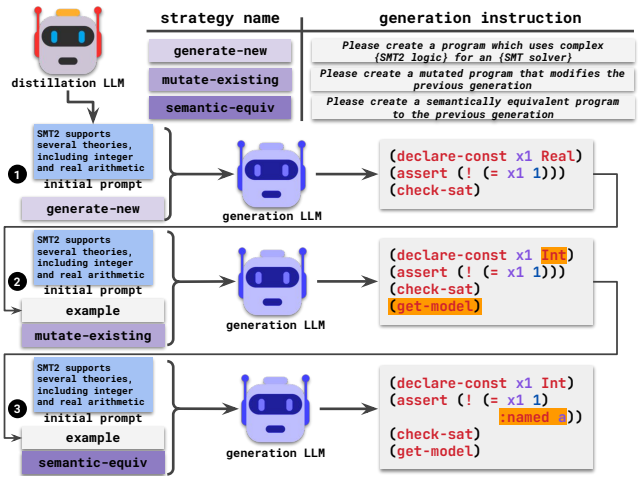


Figure 3: Fuzzing strategies and example of fuzzing loop.

3.2.1 Fuzzing Loop Algorithm. Algorithm 2 describes the fuzzing loop. The inputs are the initial input prompt and the fuzzing budget. The final output is a set of bugs identified by the user-defined oracle. First, the algorithm initializes the generation strategies (`generate-new`, `mutate-existing`, and `semantic-equiv`), which will be used to modify the input prompt during the fuzzing loop (line 2). Figure 3 (top-right) lists our three generation strategies along with their corresponding instructions. For the first invocation of the generation LLM, denoted with $\mathcal{M}_{\mathcal{G}}$, the algorithm does not yet have any examples of fuzzing inputs. Hence, it appends to the input prompt the `generate-new` generation instruction, which guides the model toward producing a first batch of fuzzing inputs (line 3).

Next, the algorithm enters the main fuzzing loop (lines 5–9), which continuously updates the prompt to create new fuzzing inputs. To this end, the algorithm selects an example from the previous batch of generated fuzzing inputs, randomly picking from all those fuzzing inputs that are valid for the SUT (line 6). In addition to the example, the algorithm also randomly picks one of the three generation strategies (line 7). The generation strategy either instructs the model to mutate the selected example (`mutate-existing`), to produce a fuzzing input that is semantically equivalent to the example (`semantic-equiv`), or to come up with a new fuzzing input (`generate-new`). The algorithm concatenates the initial input prompt, the selected example, and the selected generation strategy into a new prompt, and then queries the generation LLM with this prompt to produce another batch of fuzzing inputs (line 8).

The main fuzzing loop is repeated until the algorithm has exhausted the fuzzing budget. For each created fuzzing input, Fuzz4All passes the input to the SUT. If the user-defined oracle identifies an unexpected behavior, e.g., a crash, then the algorithm adds a report to the set of detected bugs (lines 4 and 9).

3.2.2 Example: Fuzzing Loop. Figure 3 illustrates how our fuzzing loop uses input examples and the generation strategies to create different fuzzing inputs. In this case, we are fuzzing an SMT solver where the inputs are logic formulas written in the SMT2 language. Initially ❶, there are no examples, and hence, the algorithm uses the `generate-new` strategy to synthesize new fuzzing inputs. Next, taking a generated, valid fuzzing input as an example, the algorithm queries the model to create a new input ❷ based on the `mutate-existing` strategy, which aims to mutate the selected example. We observe that the new fuzzing input subtly modifies the previous input by swapping the type of a variable as well as adding some computation. In the next fuzzing iteration ❸, the algorithm selects the previously generated fuzzing input as the example and uses the `semantic-equiv` generation strategy, which aims to create an input that does not modify the semantics of the given example. This time, we observe that the new fuzzing input simply adds a syntax tag to the selected example. In fact, the combination of generation strategies shown in the example helps Fuzz4All to generate a fuzzing input that causes an unexpected crash in the SMT solver. The crash exposes one of the real-world bugs detected by Fuzz4All during our evaluation, which has been confirmed and fixed by developers.

3.2.3 Oracle. The fuzzing inputs produced by Fuzz4All during the fuzzing loop can be used to check the behavior of the SUT against an oracle to detect bugs. The oracle is custom for each SUT, and it can be fully defined and customized by the user. For example, when fuzzing C compilers, a user could define a differential testing oracle that compares the compiler behavior under different optimization levels [81]. In this paper, we focus on simple and easy-to-define oracles, such as crashes due to segmentation faults and internal assertion failures, with more details discussed in Section 4.2.

4 EXPERIMENTAL DESIGN

We evaluate Fuzz4All on the following research questions:

- **RQ1:** How does Fuzz4All compare against existing fuzzers?
- **RQ2:** How effective is Fuzz4All in performing targeted fuzzing?
- **RQ3:** How do different components contribute to Fuzz4All's effectiveness?
- **RQ4:** What real-world bugs does Fuzz4All find?

4.1 Implementation

Fuzz4All is primarily implemented in Python. The autoprompting and fuzzing loop components of Fuzz4All contain only 872 LoC. Compared to traditional fuzzers, such as Csmith (>80K LoC), which need high manual effort to implement generators, Fuzz4All has a very lightweight implementation. Fuzz4All uses GPT4 [55] as the distillation LLM to perform autoprompting since this model is the state-of-the-art for a wide range of NLP-based reasoning tasks [9]. Specifically, we use the `gpt-4-0613` checkpoint with `max_token` of 500 provided via the OpenAI API [27]. `max_token` forces the prompts to always fit within the context window of the generation LLM. For autoprompting, we sample four candidate prompts, generate 30 fuzzing inputs each, and evaluate using a scoring function based on validity rate (as described in Section 3.1.1). For the fuzzing loop, we use the Hugging Face implementation of the StarCoder [41] model as the generation LLM, which is trained on over one trillion code tokens across over 80 languages. Our default setting when generating fuzzing inputs uses a temperature of 1, a batch size of 30, a maximum output length of 1,024 using nucleus sampling [32] with a top-p of 1.

Table 1: SUTs and baseline tools.

Language	SUT(s)	Baseline tool(s)	Version
C	GCC, Clang	GrayC [21], Csmith [81]	GCC-13.1.1
C++	G++, Clang++	YARPGen [49]	G++-13.1.1
SMT2	Z3, CVC5	TypeFuzz [59]	CVC5-1.0.5
Go	Go	go-fuzz [25]	go-1.20.6
Java	javac	Hephaestus [11]	OpenJDK-javac-18
Python	Qiskit	MorphQ [58]	qiskit-0.43.1

4.2 Systems Under Test and Baselines

To demonstrate the generality of Fuzz4All, we evaluate it on six input languages and nine SUTs. Table 1 shows each of the languages, SUTs, and the corresponding baseline tools. Note that we compare coverage on one SUT per language, with the SUT versions used for coverage measurements shown in the last column of Table 1. Except for the coverage experiments, we perform fuzzing on the nightly release of each target. Unless otherwise mentioned, we use unexpected compiler crashes as the oracle and consider a fuzzing input as valid if it compiles successfully. Each baseline fuzzer is run with its default settings. For baseline fuzzers that require input seeds, we use the default seed corpus provided in their replication repository. We now present more evaluation details for each SUT.

4.2.1 C/C++ Compilers. We target the popular GCC and Clang compilers and provide the standard C library documentation as user input to Fuzz4All by default. Our baselines include Csmith [81], a classic generation-based C compiler fuzzer, and GrayC [21], a recent mutation-based fuzzer that uses coverage feedback together with specialized mutation operators. For C++, we target new C++23 features by providing the C++23 standard documentation as input to Fuzz4All. Our baseline is YARPGen [49], a generation-based fuzzer that extends Csmith with new language features in C++ and generation policies to trigger different compiler optimizations.

4.2.2 SMT Solvers. We run Fuzz4All on Z3 and CVC5 with commonly enabled developer settings, such as `debug` and `assertion`, following prior work [59, 77, 78]. Fuzz4All generates SMT formulas as fuzzing inputs using an overview documentation of the SMT2 language and SMT solver as input by default. A fuzzing input is considered valid if the SMT solver returns either `SAT` or `UNSAT` without any error. Our baseline is state-of-the-art TypeFuzz [59], which mutates existing SMT expressions based on newly generated expressions of the same type.

4.2.3 Go Toolchain. We run Fuzz4All on the most recent version of Go. By default, we use the Go standard library documentation as input to Fuzz4All. As a baseline, we use go-fuzz [25], a coverage-guided, mutation-based fuzzer designed for Go, which generates inputs for various Go standard libraries using handwritten templates.

4.2.4 Java Compiler. We evaluate Fuzz4All on the OpenJDK Java compiler, javac, which compiles source code into bytecode. Our default input is the latest standard Java API documentation page. We compare against Hephaestus [11], a recent combined generation- and mutation-based fuzzer designed for JVM compilers and targeting type-related bugs.

4.2.5 Quantum Computing Platform. We target Qiskit [1], a popular quantum computing framework [23]. Qiskit is built on top of Python, i.e., both the input program and the compilation are defined in Python code. Thus, creating a valid input for Qiskit means using the Qiskit Python APIs in a meaningful way, e.g., to create a quantum circuit. It is challenging for traditional synthesis tools to handle dynamically typed general-purpose languages (like Python) [29, 67], not to mention the additional API constraints and quantum-specific nature of many bugs [57], making fuzzing Qiskit a particularly difficult challenge. Our baseline is MorphQ [58], a recent fuzzer that uses a template- and grammar-based approach to generate valid quantum programs and then applies metamorphic transformations.

Unlike for the other SUTs, which receive fuzzing inputs in a file, to invoke Qiskit, we must run the generated Python program itself. As an oracle, we add statements at the end of the generated Python file, which collect all `QuantumCircuit` objects via Python's built-in introspection APIs and then apply two oracles on each circuit. The two oracles are directly borrowed from previous work for a fair comparison [58]. The first oracle compiles the circuit via a `transpile` call with different optimization levels and reports any crash. The second oracle converts the circuit to its lower-level QASM [16] representation and then reads it back, reporting any crash.

4.3 Experimental Setup and Metrics

Fuzzing campaigns. For RQ1, we use a fuzzing budget of 24 hours (including autoprompting), which is used commonly in prior work [36]. To account for variance, we repeat the experiment for both Fuzz4All and the baselines five times. Due to the high cost of experiments, for later RQs, we use a fuzzing budget of 10,000 generated fuzzing inputs and repeat four times for the ablation study.

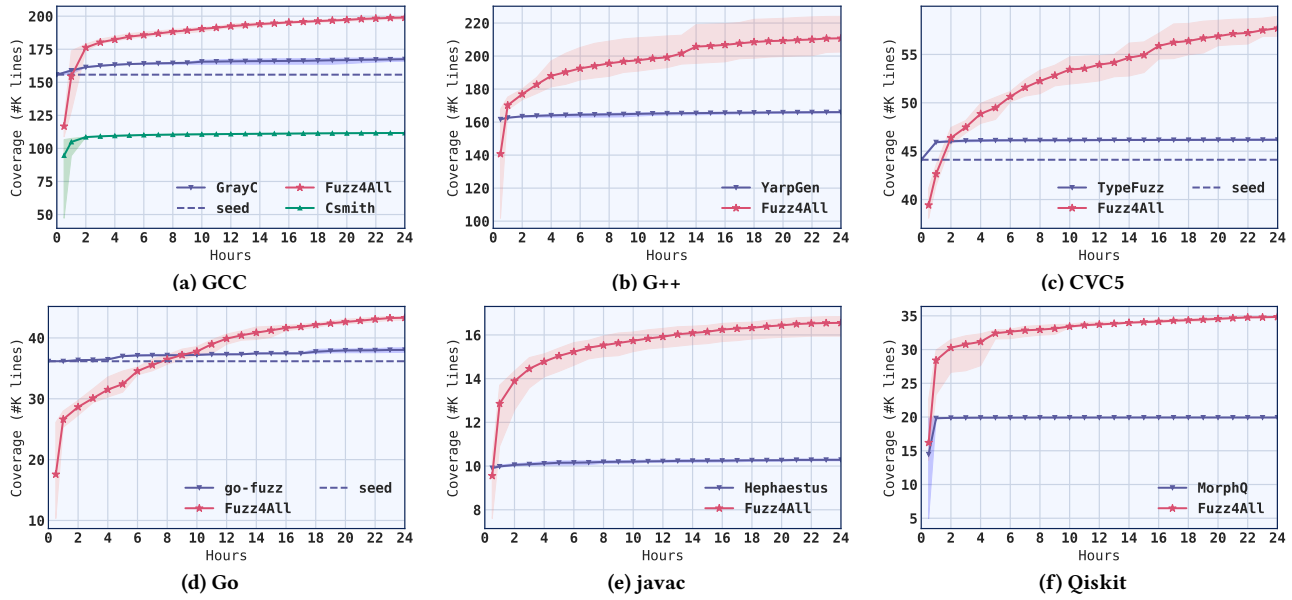
Environment. Experiments are conducted on a 64-core workstation with 256 GB RAM running Ubuntu 20.04.5 LTS with 4 NVIDIA RTX A6000 GPUs (only one GPU is used per fuzzing run).

Metrics. We use the widely adopted measure of *code coverage* for evaluating fuzzing tools [7, 36, 76]. To be uniform, we report the line coverage for each of the targets studied in the evaluation. Following prior work [36], we use the Mann-Whitney U-test [52] to compute statistical significance and indicate significant ($p < 0.05$) coverage results in applicable tables (Tables 2 and 4) with *. We additionally measure the *validity rate* (% valid) of inputs as the percentage of fuzzing inputs generated that are valid and unique. As Fuzz4All supports both general and targeted fuzzing, to assess the effectiveness of targeted fuzzing, we report the *hit rate*, i.e., the percentage of fuzzing inputs that use a specific target feature (checked with simple regular expressions). Finally, we also report the most important metric and goal of fuzzing: the number of bugs detected by Fuzz4All for each of our nine SUTs.

5 RESULTS

5.1 RQ1: Comparison against Existing Fuzzers

5.1.1 Coverage over Time. Figure 4 shows the 24-hour coverage trend of Fuzz4All compared with the baselines, where the solid line shows average coverage and the area indicates the minimum



(a) GCC (b) G++ (c) CVC5

(d) Go (e) javac (f) Qiskit

Figure 4: Coverage trend of Fuzz4All against state-of-the-art fuzzers in a 24-hour fuzzing campaign.

Table 2: Fuzz4All against state-of-the-art fuzzers (* indicates statistically significant coverage improvement).

Target	Fuzzer	# programs	% valid	Coverage	
GCC	GrayC	104,326	95.96%	167,453	
	Csmith	61,883	99.99%	111,668	
	Fuzz4All	44,324	37.26%	***198,927**	+18.8%
G++	YARPGen	255,581	99.99%	166,614	
	Fuzz4All	26,365	40.74%	***210,743**	+26.5%
CVC5	TypeFuzz	43,001	93.24%	46,174	
	Fuzz4All	36,054	47.63%	***57,674**	+24.9%
Go	go-fuzz	20,002	100.00%	38,024	
	Fuzz4All	22,817	23.02%	***43,317**	+13.7%
javac	Hephaestus	728,217	57.22%	10,285	
	Fuzz4All	31,967	49.05%	***16,552**	+60.9%
Qiskit	MorphQ	38,474	100.00%	19,929	
	Fuzz4All	33,454	24.90%	***34,988**	+75.6%

and maximum across five runs. We observe that Fuzz4All achieves the highest coverage by the end of the fuzzing campaign across all targets, with an average improvement of 36.8% compared to the top performing baselines. Contrasting with generation-based fuzzers (i.e., YARPGen and MorphQ), Fuzz4All is able to almost immediately achieve higher coverage, demonstrating the powerful generative ability of LLMs in producing diverse code snippets compared to traditional program generation techniques. While mutation-based fuzzers (i.e., go-fuzz and GrayC) are able to achieve higher coverage in the beginning through the use of high quality seeds, the coverage gained via mutations rapidly falls off and Fuzz4All is able to slowly but surely cover more code. Note that we include the autoprompting time as part of the fuzzing budget for a fair comparison, which incurs negligible overhead (avg. 2.3 minutes per fuzzing campaign).

Unlike the baseline fuzzers, which reach a coverage plateau by the end of the 24-hour period, Fuzz4All keeps finding inputs that cover new code, even near the end of the fuzzing campaign. Recall that during each iteration of Fuzz4All's fuzzing loop, the original input prompt is updated with both a new example and a generation strategy (Section 3.2), nudging the LLM to generate new fuzzing inputs. We hypothesize that this allows Fuzz4All to effectively generate new and diverse fuzzing inputs even after a long period of fuzzing, leading to sustained coverage increase.

5.1.2 Generation Validity, Number, and Coverage. We examine the number of fuzzing inputs generated and their validity rate across our studied SUTs. In Table 2, Column "# programs" represents the number of unique inputs generated, "% valid" is the percentage of fuzzing inputs that are valid, and "Coverage" shows the final coverage obtained by each fuzzer along with the relative improvement over the best baseline. We first observe that almost all traditional fuzzing tools can achieve a very high validity rate apart from Hephaestus, which purposefully generates invalid code (focused on incorrect types) to check for miscompilation bugs. In contrast, Fuzz4All has a lower percentage of valid fuzzing inputs generated (56.0% average reduction compared to baseline tools). Furthermore, the raw number of fuzzing inputs generated by baseline tools are also much higher. By using an LLM as the generation engine, Fuzz4All is bottlenecked by GPU inference, leading to 43.0% fewer fuzzing inputs compared to traditional fuzzers.

In spite of the lower validity rate and number of fuzzing inputs, Fuzz4All generates much more diverse programs compared to traditional fuzzing tools, as evidenced by the high coverage obtained (+36.8% average increase). Additionally, even invalid code snippets that are close to valid can be useful for fuzzing, as they allow for finding bugs in the validation logic of the SUT. In Section 5.4, we further describe the various types of bugs detected by Fuzz4All,

with both valid and invalid code snippets, to additionally showcase the benefit of generating diverse fuzzing inputs.

We note that Fuzz4All achieves a wide range of validity rates and numbers of fuzzing inputs across different SUTs. The number of fuzzing inputs varies across targets due to the varying cost to invoke the SUT after each fuzzing iteration for bug detection. Regarding validity rate, a general-purpose programming language, such as C, has a relatively lower validity rate compared to domain-specific languages, such as the SMT2 language used for SMT solvers. A more rigorous language, e.g., Go, which does not allow any declared but unused variables, has an even lower validity rate. We also observe a low validity rate for fuzzing quantum computing platforms. As quantum computing is an emerging area with its own set of library APIs, the generation LLM may not have seen as many examples of quantum programs during its training as for more established languages. Nevertheless, Fuzz4All is still able to leverage user-provided documentation to generate interesting fuzzing inputs that use quantum library APIs and achieve an impressive coverage improvement (+75.6%) compared to the state-of-the-art fuzzer.

5.2 RQ2: Effectiveness of Targeted Fuzzing

We now evaluate the ability of Fuzz4All to perform targeted fuzzing, i.e., to generate fuzzing inputs that focus on a particular feature. For each target SUT and language, we target three different example features and compare them to the setup with general user input, as used for RQ1 (described in Section 4.3). These features are built-in libraries or functions/APIs (Go, C++ and Qiskit), language keywords (C and Java), and theories (SMT). The user input for the targeted fuzzing runs is documentation of the particular feature we are focusing on. Table 3 shows the results of targeted fuzzing as well as the default general fuzzing used in RQ1. Each column represents a targeted fuzzing run where we focus on one feature. The value in each cell shows the hit rate of the feature (Section 4.3) for a particular fuzzing run. We also include the coverage results obtained.

We observe that targeting a specific feature yields a high amount of fuzzing inputs that directly use the feature, with an average hit rate of 83.0%. This result demonstrates that Fuzz4All indeed performs targeted fuzzing by prompting the generation LLM with an input prompt that describes a particular feature. Furthermore, we observe that fuzzing on features that are related can lead to a moderately high cross-feature hit rate (i.e., hit rate of feature `X` on fuzzing run for feature `Y`). For example, the C keywords `typedef` and `union` are both related to type operations, and hence, their cross-feature hit rate is high compared to an unrelated feature, such as `goto`. As shown in Table 3, a general fuzzing approach, while achieving the highest overall code coverage, can be extremely inefficient in targeting a specific feature (average 96.0% reduction in hit rate compared with Fuzz4All's targeted fuzzing). For example, in Qiskit, the general fuzzing campaign has a 0% hit rate of the three target features. This can be explained by the fact that these features were added recently to Qiskit and are not yet widely used, thus being extremely rare in the LLM training data. However, by providing suitable user input during the targeted fuzzing campaign, Fuzz4All can successfully generate fuzzing inputs that use these new features. This ability of Fuzz4All will be valuable to developers who want to test novel features or components of a SUT.

Table 3: Hit rate and coverage during targeted fuzzing.

	C targeted campaign (keywords)			
	`typedef`	`union`	`goto`	General
Hit rate `typedef`	**83.11%**	47.16%	0.48%	4.38%
Hit rate `union`	10.80%	**80.43%**	0.10%	0.32%
Hit rate `goto`	0.22%	0.11%	**77.62%**	1.16%
Coverage	123,226	125,041	120,452	188,148
	C++ targeted campaign (built-in functions)			
	`apply`	`expected`	`variant`	General
Hit rate `apply`	**70.23%**	0.41%	0.68%	0.32%
Hit rate `expected`	0.26%	**79.72%**	0.94%	1.33%
Hit rate `variant`	1.16%	5.98%	**93.19%**	3.63%
Coverage	182,261	175,963	182,333	193,254
	SMT targeted campaign (theories)			
	`Array`	`BitVec`	`Real`	General
Hit rate `Array`	**82.23%**	2.08%	1.44%	11.07%
Hit rate `BitVec`	2.57%	**88.48%**	0.86%	5.46%
Hit rate `Real`	1.45%	0.17%	**96.01%**	17.36%
Coverage	46,392	48,841	47,619	52,449
	Go targeted campaign (built-in libraries)			
	`atomic`	`atomic`	`heap`	General
Hit rate `atomic`	**90.09%**	0.04%	0.06%	1.01%
Hit rate `big`	0.18%	**97.20%**	0.23%	3.63%
Hit rate `heap`	0.30%	0.04%	**91.18%**	2.22%
Coverage	10,156	12,986	9,790	37,561
	Java targeted campaign (keywords)			
	`instanceof`	`synchronized`	`finally`	General
Hit rate `instanceof`	**88.00%**	0.08%	0.85%	1.86%
Hit rate `synchronized`	0.16%	**94.80%**	0.16%	0.85%
Hit rate `finally`	0.51%	3.17%	**78.62%**	0.82%
Coverage	14,546	13,972	13,203	16,128
	Qiskit targeted campaign (APIs)			
	`switch`	`for loop`	`linear`	General
Hit rate `switch`	**71.76%**	0.00%	0.00%	0.00%
Hit rate `for loop`	0.17%	**75.97%**	0.00%	0.00%
Hit rate `linear`	0.00%	0.00%	**54.79%**	0.00%
Coverage	30,597	26,703	29,535	33,853

5.3 RQ3: Ablation Study

To study how each component of Fuzz4All contributes to the overall fuzzing effectiveness, we conduct an ablation study based on the two key components of Fuzz4All: (a) Autoprompting, the type of initial input prompt provided to the generation LLM; (b) Fuzzing loop, the use of selected examples and generation strategies. We study three variants for each of the two key components. Table 4 shows the coverage and validity rate of our studied variants.

5.3.1 Autoprompting. First, we examine the effect of different initial inputs provided to the generation LLM. To reduce the impact of additional factors, we fix the generation strategy to only use `generate-new` and study three variants[1]: 1) `no input` does not use any initial prompts 2) `raw prompt` directly uses the raw user input as

[1]The impact of additional generation strategies can be found in Section 5.3.2.

Table 4: Effectiveness of variants (* indicates statistically significant coverage improvement compared w/ 2nd best variant).

	Variants	Description	C		C++		SMT		Go		Java		Qiskit	
			Cov.	% valid	Cov.	% valid	Cov.	% valid	Cov.	% valid	Cov.	% valid	Cov.	% valid
Auto prompt.	`no input`	no initial prompt	127,261	42.57%	181,493	51.63%	50,838	49.49%	35,765	39.54%	14,374	50.25%	31,701	34.63%
	`raw prompt`	use user-provided input	137,204	33.95%	189,030	33.79%	49,697	39.49%	36,168	16.84%	15,445	37.64%	31,922	22.74%
	`autoprompt`	apply autoprompting	182,530	39.09%	190,318	36.62%	51,496	45.04%	36,732	24.87%	15,838	45.54%	32,691	29.12%
Fuzzing loop	`w/o example`	`generate-new` w/o example	143,349	34.23%	190,288	28.25%	50,089	18.41%	35,839	19.38%	15,444	44.69%	32,663	24.04%
	`w/ example`	`generate-new` w/ example	182,530	39.09%	190,318	36.62%	51,496	45.04%	36,732	24.87%	15,838	45.54%	32,691	29.12%
	Fuzz4All	all strategies w/ example	**185,491**	40.58%	***193,845**	41.22%	***53,069**	50.06%	***37,981**	32.00%	***16,209**	50.99%	***33,913**	27.45%

the initial prompt, 3) `autoprompt` applies autoprompting to generate the initial prompt. We observe that across all studied languages, the `no input` variant achieves the lowest coverage. In `no input`, we do not provide any initial prompt, which provides useful information on the features we want to generate fuzzing inputs for. As such, the LLM can only generate simple code snippets with high validity rate but is less effective in covering the SUT. We observe a coverage boost as we use the `raw prompt` variant, where we provide the raw documentation as the initial prompt. However, we can further improve both the code coverage and the validity rate by using our autoprompting stage to distill the user input into a concise but informative prompt (`autoprompt`), instead of using the raw user input. Directly using the user-provided input may include information that is irrelevant for fuzzing, leading to both a lower validity rate (as the generation LLM may struggle to understand the raw documentation) and lower coverage (since, unlike our autoprompting generated prompt, the raw documentation is not designed to be used for LLM generation).

5.3.2 Fuzzing loop. Next, we examine the different variants of our fuzzing loop setup by keeping the initial prompt the same (by using the default autoprompting): 1) `w/o example` does not select an example during the fuzzing loop (i.e., it continuously samples from the same initial prompt), 2) `w/ example` selects an example but only uses the `generate-new` instruction[2], 3) Fuzz4All is the full approach with all generation strategies used. We first observe that by only sampling from the same input (`w/o example`), LLMs will often repeatedly generate the same or similar fuzzing inputs. On average, 8.0% of the fuzzing inputs generated are repeated in `w/o example` compared to only 4.7% when using the full Fuzz4All approach. Adding an example to the input prompt (`w/ example`) avoids sampling from the same distribution and improves both the coverage and the validity rate. Finally, the full Fuzz4All approach achieves the highest coverage across all SUTs. Compared to the `w/ example` variant (the second-best), the full Fuzz4All adds additional generation strategies, `semantic-equiv` and `mutate-existing`, which provide useful instructions to the generation LLM.

5.4 RQ4: Bug Finding

Table 5 summarizes the bugs found by Fuzz4All on our nine studied SUTs. In total, Fuzz4All detects 98 bugs, with 64 bugs already confirmed by the developers as previously unknown. These results not only demonstrate the practical effectiveness of Fuzz4All in finding large amounts of bugs but also the promised generality of Fuzz4All across languages and SUTs. A detailed list of reported bugs and issue links can be found in our artifact.

[2]Note that `autoprompt` and `w/ example` are the same variant, but we include them separately for ease of comparison.

Table 5: Summary of Fuzz4All-detected bugs.

	Total	Confirmed		Pending	Won't fix
		Unknown	Known		
GCC	30	14	11	5	0
Clang	27	18	9	0	0
CVC5	9	7	2	0	0
Z3	14	12	0	0	2
Go	4	2	2	0	0
Java	3	3	0	0	0
Qiskit	11	8	2	1	0
Total	98	64	26	6	2

```
#include <optional>
void y(std::optional<int> z)
        noexcept(noexcept(std::optional<int>{z})) {}
```

(a) GCC bug: Internal compiler error (segmentation fault)

```
#include <iostream>
using E = std::numeric_limits<int>;
auto fail(E e) -> decltype(throw e, void()) { throw e; }
```

(b) Clang bug: Segmentation fault

```
package main
import ("runtime")
func main() { runtime.ReadMemStats(nil) }
```

(c) Go bug: Segmentation violation

```
from qiskit import QuantumCircuit, ClassicalRegister
crz = ClassicalRegister(1, name="crz")
qc = QuantumCircuit(crz)
qc.qasm(filename="my.qasm")
QuantumCircuit.from_qasm_file("my.qasm")
```

(d) Qiskit bug: Crash

Figure 5: Exemplary bugs found by Fuzz4All.

5.4.1 Examples. Figure 5a shows a bug found in GCC when using `noexcept(x)`, a C++ feature that specifies a function is non-throwing if `x` evaluates to `true`. In this example bug, Fuzz4All generates a rather complex code using `std::optional`, which indicates that a particular value may or may not be present at runtime. While this code is valid and should compile correctly, this combination of difficult runtime dependencies cause GCC to crash with an `internal compiler error`. We note that this bug cannot be found by prior techniques since they simply do not support the `noexcept` feature.

The developers have already confirmed and fixed this bug. Interestingly, they even added a slightly modified version of our submitted code snippet to the official test suite of GCC.

Figure 5b shows a bug found in Clang, where the invalid code leads to a segmentation fault. Fuzz4All uses an unusual syntax for function declaration (i.e., `auto x (...) -> return_type`), which makes use of the `decltype` operation in C++. However, the bug occurs when the `throw` statement inside of the `decltype` is evaluated first, skipping the evaluation of the return type since `throw` exits the scope early and crashes Clang. This code, while invalid, is still useful to reveal a bug in the Clang frontend as confirmed by the developers. Additionally, prior fuzzing tools can hardly find this bug since they typically focus on generating valid code only and do not handle the especially difficult-to-model `decltype` function.

Figure 5c shows a bug found in Go where a `nil` input causes a segmentation fault instead of producing a useful failure message. This bug is found by targeting the `runtime` Go standard library, where we provide the documentation, which includes the description of the `ReadMemStats` function. The bug has been confirmed and fixed by the developers. While this bug might look simple (invoking a singular function), it cannot be found by the go-fuzz baseline simply because go-fuzz requires manually written templates to target specific libraries, and `runtime` is not a part of any such template. With Fuzz4All, users can directly target any Go standard libraries by providing relevant input information (e.g., documentation).

Figure 5d shows a bug found in Qiskit's QASM exporter. A quantum program, represented by the `qc` variable, is exported to QASM, a low level representation, silently generating an invalid output file, which leads to a crash when being reimported. The problem is that the exporter represents the register in QASM using its name as identifier, i.e.,`"crz"`, which also is the name of a well-known operation of the QASM language, thus making the generated code ambiguous. Note that prior work [58] could not find this bug because they use pre-defined templates with only anonymous registers, whereas Fuzz4All effectively leverages the quantum knowledge of LLMs to inject a meaningful string literal for detecting this bug.

6 THREATS TO VALIDITY

Internal. The main internal threat comes from the implementation of Fuzz4All. To address this, we performed code reviews and testing to ensure correctness. Furthermore, we run each baseline from their provided replication package whenever possible.

External. The main external threat is our evaluation targets. To support our generality claim, we apply Fuzz4All on nine different SUTs across six languages. Additionally, to account for variance in long fuzzing runs, we repeat the 24-hour fuzzing campaign five times and check for statistically significant results. Since the generation LLM leverages the knowledge acquired during its training done within the last year, reapplying Fuzz4All using the exact checkpoint of the LLM (StarCoder) used in this work might degrade the effectiveness in the future due to data-shift. Fuzz4All can mitigate this using the autoprompting step where more up-to-date documentation/example code allows the model to also generate up-to-date fuzzing inputs. One additional threat comes from the use of the distillation LLM to generate the initial inputs, where the LLM may "hallucinate", i.e., produce made-up or inaccurate information [30] . This limitation is common to most pipelines that use LLMs, and we hope to address it in our future work.

7 CONCLUSION

We present Fuzz4All, a universal fuzzer leveraging LLMs to support both general and targeted fuzzing of arbitrary SUTs that take in a multitude of programming languages. Fuzz4All uses a novel autoprompting stage to produce input prompts that concisely summarize the user-provided inputs. In its fuzzing loop, Fuzz4All iteratively updates the initial input prompt with both code examples and generation strategies aimed at producing diverse fuzzing inputs. Evaluation results on nine different SUTs across six different languages demonstrate that Fuzz4All is able to significantly improve coverage compared to state-of-the-art tools. Furthermore, Fuzz4All is able to detect 98 bugs with 64 already confirmed by developers as previously unknown.

DATA AVAILABILITY

Our code and data are available at: https://doi.org/10.5281/zenodo.10456883 and https://github.com/fuzz4all/fuzz4all

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